Coffee consumption and risk of fractures: a systematic review and dose-response meta-analysis.

PURPOSE: The data on the association between coffee consumption and the risk of fractures are inconclusive. We performed a comprehensive literature review and meta-analysis to better quantify this association. METHODS: We identified all potentially relevant articles by searching MEDLINE, EMBASE, Cochrane Library, Web of Science, SCOPUS, and CINAHL (until February 2013). The keywords "coffee," "caffeine," "drink," and "beverage" were used as the exposure factors, and the keyword "fracture" was used as the outcome factor. We determined the overall relative risk (RR) and confidence interval (CI) for the highest and lowest levels of coffee consumption. A dose-response analysis was performed to assess the risk of fractures based on the level of coffee consumption. RESULTS: We included 253,514 participants with 12,939 fracture cases from 9 cohort and 6 case-control studies. The estimated RR of fractures at the highest level of coffee consumption was 1.14 (95% CI: 1.05-1.24; I(2)=0.0%) in women and 0.76 (95% CI: 0.62-0.94; I(2)=7.3%) in men. In the dose-response analysis, the pooled RRs of fractures in women who consumed 2 and 8 cups of coffee per day were 1.02 (95% CI: 1.01-1.04) and 1.54 (95% CI: 1.19-1.99), respectively. CONCLUSIONS: Our meta-analysis suggests that daily consumption of coffee is associated with an increased risk of fractures in women and a contrasting decreased risk in men. However, future well-designed studies should be performed to confirm these findings.